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JEE MAIN · QUESTION PAPER

JEE Main 2023 (Online) 31st January Morning Shift

JEE Main Question Paper with Answer Key

31 Jan 2023	90	16	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	24 questions
Mathematics	23 questions
Physics	27 questions
Out of syllabus	16 questions
Answer key	at the end

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Chemistry 24 questions

Q1. MCQ (Single Correct Answer)

Open solution

When Cu^{2+} ion is treated with KI, a white precipitate, X appears in solution. The solution is titrated with sodium thiosulphate, the compound Y is formed. X and Y respectively are :

(A) $X = CuI_2$ | $Y = Na_2S_4O_6$

(B) $X = Cu_2I_2$ | $Y = Na_2S_4O_6$

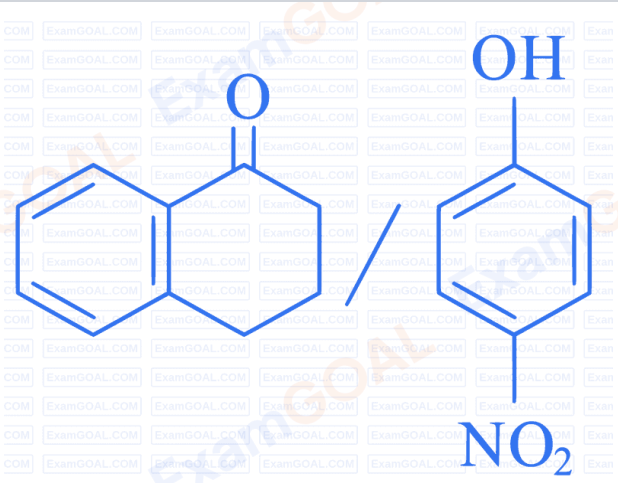
(C) $X = CuI_2$ | $Y = Na_2S_2O_3$

(D) $X = Cu_2I_2$ | $Y = Na_2S_4O_5$

Q2. MCQ (Single Correct Answer)

Open solution

Match items of column I and II

	Column I (Mixture of compounds)		Column II (Separation Technique)
A.	H_2O/CH_2Cl_2	i.	Crystallization
B.		ii.	Differential solvent extraction
C.	Kerosene / Naphthalene	iii.	Column chromatography
D.	$C_6H_{12}O_6/NaCl$	iv.	Fractional Distillation

Correct match is

(A) A-(ii), B-(iv), C-(i), D-(iii)

(B) A-(iii), B-(iv), C-(ii), D-(i)

(C) A-(i), B-(iii), C-(ii), D-(iv)

(D) A-(ii), B-(iii), C-(iv), D-(i)

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Q3. MCQ (Single Correct Answer)

Open solution

Consider the following reaction

$$\text{Propanal} + \text{Methanal} \xrightarrow[\text{(ii) } \Delta \\ \text{(iii) NaCN} \\ \text{(iv) } H_3O^+]{\text{(i) dil. NaOH}} \text{Product B } (C_5H_8O_3)$$

The correct statement for product B is: it is

(A) racemic mixture and is neutral

(B) optically active and adds one mole of bromine

(C) racemic mixture and gives a gas with saturated $NaHCO_3$ solution

(D) optically active alcohol and is neutral

Q4. MCQ (Single Correct Answer)

Open solution

Nd^{2+} = _____

(A) $4f^4 6s^2$ (B) $4f^3$ (C) $4f^4$ (D) $4f^2 6s^2$

Q5. MCQ (Single Correct Answer)

Open solution

The correct order of basicity of oxides of vanadium is :

(A) $V_2O_3 > V_2O_5 > V_2O_4$ (B) $V_2O_3 > V_2O_4 > V_2O_5$

(C) $V_2O_5 > V_2O_4 > V_2O_3$ (D) $V_2O_4 > V_2O_3 > V_2O_5$

Q6. MCQ (Single Correct Answer)

Open solution

The correct increasing order of the ionic radii is

(A) $Cl^- < Ca^{2+} < K^+ < S^{2-}$ (B) $K^+ < S^{2-} < Ca^{2+} < Cl^-$

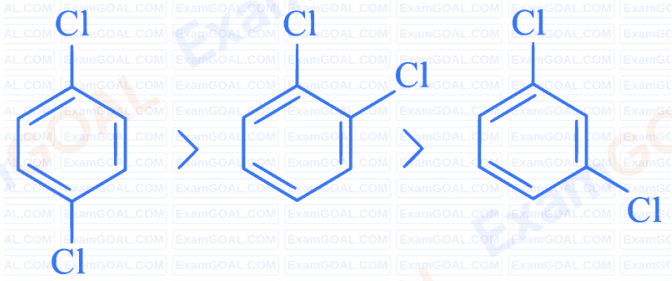
(C) $Ca^{2+} < K^+ < Cl^- < S^{2-}$ (D) $S^{2-} < Cl^- < Ca^{2+} < K^+$

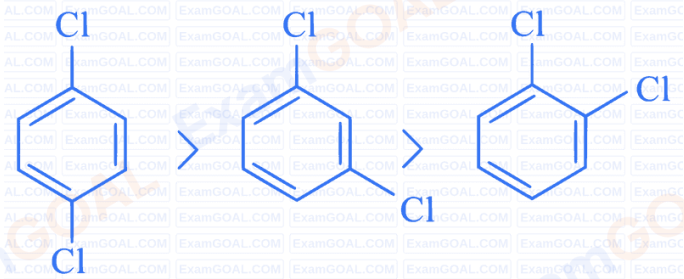
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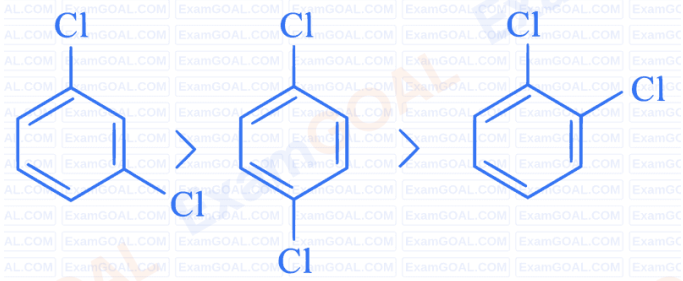
Q7. MCQ (Single Correct Answer)

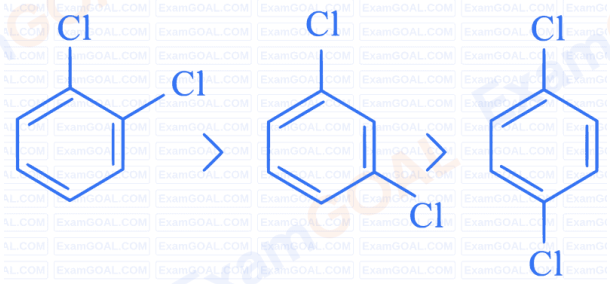
Open solution

The correct order of melting points of dichlorobenzenes is

(A) 

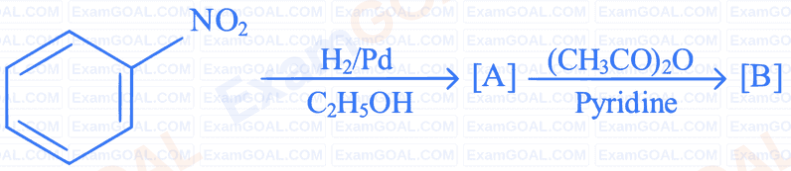
(B) 

(C) 

(D) 

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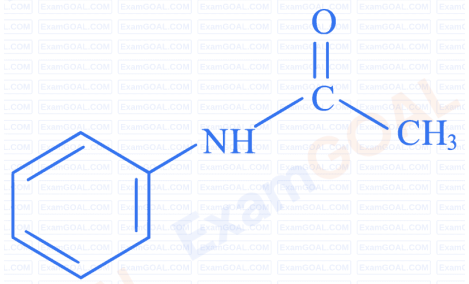
Q8. MCQ (Single Correct Answer)



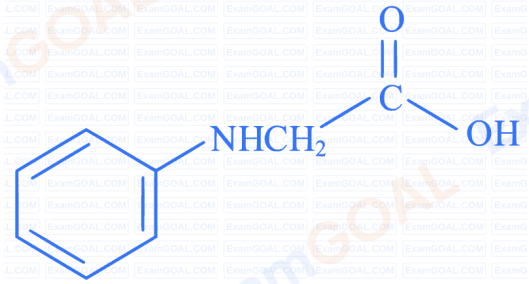
Open solution

Consider the above reaction and identify the product B.

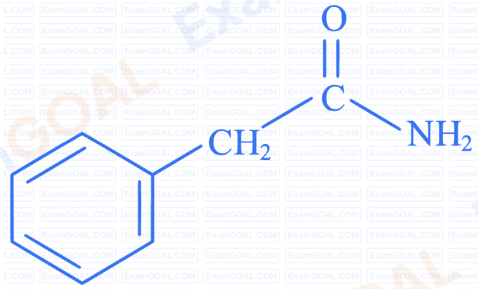
(A)



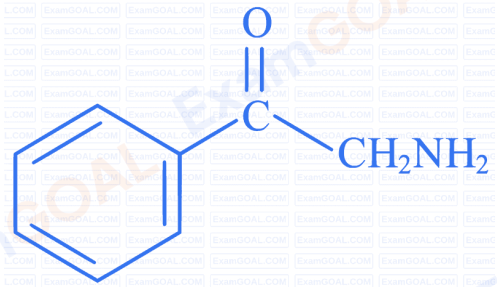
(B)



(C)



(D)



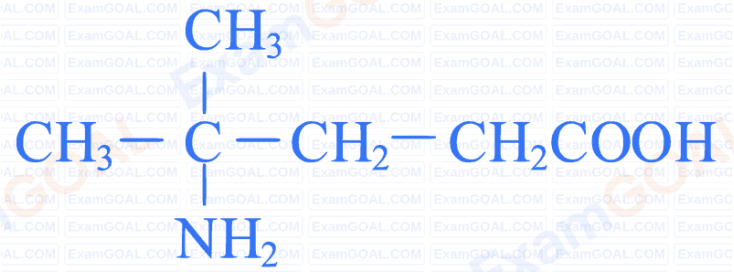
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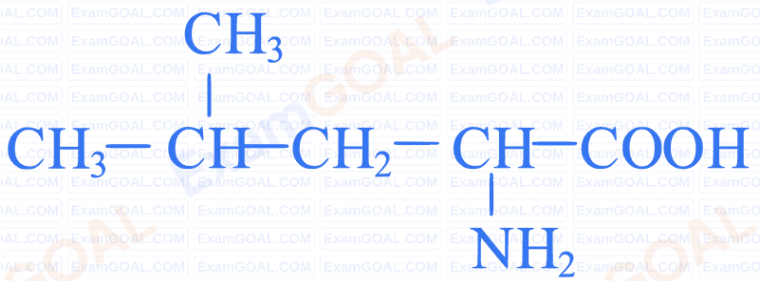
Q9. MCQ (Single Correct Answer)

A protein 'X' with molecular weight of $70,000\ \mathrm{u}$, on hydrolysis gives amino acids. One of these amino acids is

Open solution

(A) 

(B) 

(C) 

(D) 

Q10. MCQ (Single Correct Answer)

Choose the correct set of reagents for the following conversion.

trans $(\mathrm{Ph-CH=CH-CH_3}) \rightarrow$ cis $(\mathrm{Ph-CH=CH-CH_3})$

Open solution

(A) $\mathrm{Br_2, alc \cdot KOH, NaNH_2, H_2}$ Lindlar Catalyst

(B) $\mathrm{Br_2, aq \cdot KOH, NaNH_2, Na}$ (Liq $\mathrm{NH_3}$)

(C) $\mathrm{Br_2, alc \cdot KOH, NaNH_2, Na}$ (Liq $\mathrm{NH_3}$)

(D) $\mathrm{Br_2, aq \cdot KOH, NaNH_2, H_2}$ Lindlar Catalyst

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Q11. MCQ (Single Correct Answer)

Which transition in the hydrogen spectrum would have the same wavelength as the Balmer type transition from $n = 4$ to $n = 2$ of He^+ spectrum

(A) $n = 3$ to $n = 4$ (B) $n = 2$ to $n = 1$ (C) $n = 1$ to $n = 2$ (D) $n = 1$ to $n = 3$

Open solution

Q12. MCQ (Single Correct Answer)

Which one of the following statements is correct for electrolysis of brine solution?

(A) O_2 is formed at cathode

(B) H_2 is formed at anode

(C) Cl_2 is formed at cathode

(D) OH^- is formed at cathode

Open solution

Q13. MCQ (Single Correct Answer)

Cobalt chloride when dissolved in water forms pink colored complex $\underline{X}$ which has octahedral geometry. This solution on treating with conc HCl forms deep blue complex, $\underline{Y}$ which has a $\underline{Z}$ geometry. X, Y and Z, respectively, are

(A) $X = [Co(H_2O)_6]^{2+}, Y = [CoCl_4]^{2-}, Z =$ Tetrahedral

(B) $X = [Co(H_2O)_6]^{2+}, Y = [CoCl_6]^{3-}, Z =$ Octahedral

(C) $X = [Co(H_2O)_4Cl_2]^{+}, Y = [CoCl_4]^{2-}, Z =$ Tetrahedral

(D) $X = [Co(H_2O)_6]^{3+}, Y = [CoCl_6]^{3-}, Z =$ Octahedral

Open solution

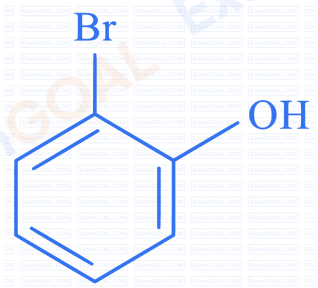
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Q14. MCQ (Single Correct Answer)

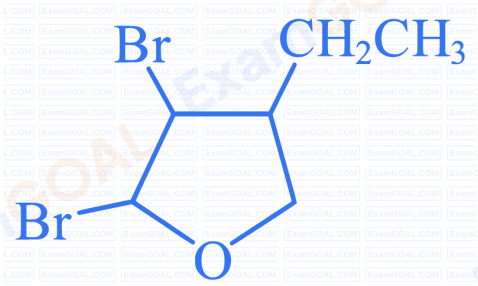
Open solution

An organic compound 'A' with empirical formula C_6H_6O gives sooty flame on burning. Its reaction with bromine solution in low polarity solvent results in high yield of B. B is

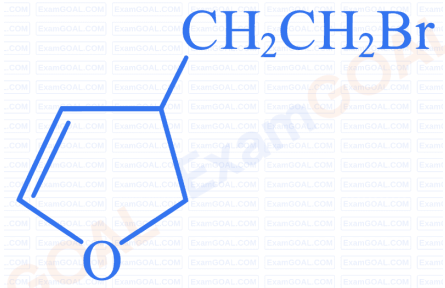
(A)



(B)



(C)



(D)



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Q15. MCQ (Single Correct Answer)

Open solution

Match List I with List II

	List I		List II
A.	XeF_4	I.	See-saw
B.	SF_4	II.	Square-planar
C.	NH_4^+	III.	Bent T-shaped
D.	BrF_3	IV.	Tetrahedral

Choose the correct answer from the options given below :

(A) A - II, B - I, C - III, D - IV

(B) A - II, B - I, C - IV, D - III

(C) A - IV, B - I, C - II, D - III

(D) A - IV, B - III, C - II, D - I

Q16. Numerical

Open solution

Zinc reacts with hydrochloric acid to give hydrogen and zinc chloride. The volume of hydrogen gas produced at STP from the reaction of $11.5\ g$ of zinc with excess HCl is _____ L (Nearest integer)

(Given : Molar mass of Zn is $65.4\ g\ mol^{-1}$ and Molar volume of H_2 at $STP = 22.7\ L$)

Q17. Numerical

Open solution

The logarithm of equilibrium constant for the reaction $Pd^{2+} + 4Cl^- \rightleftharpoons PdCl_4^{2-}$ is _____ (Nearest integer)

Given : $\frac{2.303RT}{F} = 0.06\ V$

$Pd^{2+}_{(aq)} + 2e^- \rightleftharpoons Pd(s) \quad E^{\ominus} = 0.83\ V$

$PdCl_4^{2-}(aq) + 2e^- \rightleftharpoons Pd(s) + 4Cl^-(aq) E^{\ominus} = 0.65\ V$

Q18. Numerical

Open solution

The enthalpy change for the conversion of $\frac{1}{2}Cl_2(g)$ to $Cl^-(aq)$ is $(-)$ _____ $kJmol^{-1}$ (Nearest integer)

Given : $\Delta_{dis}H^{\ominus}_{Cl^{\theta}_{2(g)}} = 240\ kJ\ mol^{-1}, \Delta_{eg}H^{\ominus}_{Cl_{(g)}} = -350\ kJ\ mol^{-1}$,

$\Delta_{hyd}H^{\Theta}_{Cl^-_{(g)}} = -380\ kJ\ mol^{-1}$

Q19. Numerical

Open solution

On complete combustion, $0.492\ g$ of an organic compound gave $0.792\ g$ of CO_2. The % of carbon in the organic compound is _____ (Nearest integer)

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Q20. Numerical

At $27^{\circ}C$, a solution containing $2.5\ g$ of solute in $250.0\ mL$ of solution exerts an osmotic pressure of $400\ Pa$. The molar mass of the solute is _____ $g\ mol^{-1}$ (Nearest integer)

(Given : $R = 0.083\ L\ bar\ K^{-1}\ mol^{-1}$)

Open solution

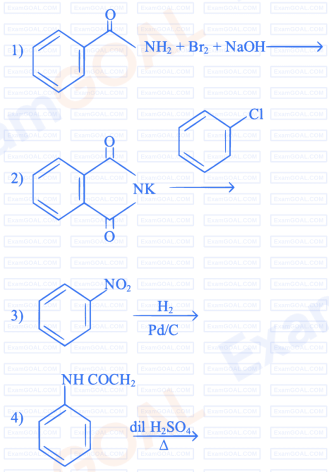
Q21. Numerical

The oxidation state of phosphorus in hypophosphoric acid is + _____.

Open solution

Q22. Numerical

How many of the transformations given below would result in aromatic amines?



Open solution

Q23. Numerical

$A \rightarrow B$

The rate constants of the above reaction at 200 K and 300 K are 0.03 min^{-1} and 0.05 min^{-1} respectively. The activation energy for the reaction is _____ J (Nearest integer)

(Given : $\ln 10 = 2.3$

$R = 8.3\ J\ K^{-1}\ mol^{-1}$

$\log 5 = 0.70$

$\log 3 = 0.48$

$\log 2 = 0.30$)

Open solution

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Q24. Numerical

For reaction : $SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$

$K_p = 2 \times 10^{12}$ at $27^\circ C$ and $1\ atm$ pressure. The K_c for the same reaction is _____ $\times 10^{13}$. (Nearest integer)

(Given $R = 0.082\ L\ atm\ K^{-1}\ mol^{-1}$)

Open solution

Mathematics 23 questions

Q25. MCQ (Single Correct Answer)

The number of real roots of the equation $\sqrt{x^2-4x+3} + \sqrt{x^2-9} = \sqrt{4x^2-14x+6}$, is :

(A) 0 (B) 1 (C) 3 (D) 2

Open solution

Q26. MCQ (Single Correct Answer)

A bag contains 6 balls. Two balls are drawn from it at random and both are found to be black. The probability that the bag contains at least 5 black balls is :

(A) $\frac{3}{7}$ (B) $\frac{5}{6}$ (C) $\frac{5}{7}$ (D) $\frac{2}{7}$

Open solution

Q27. MCQ (Single Correct Answer)

If $\sin^{-1}\frac{\alpha}{17} + \cos^{-1}\frac{4}{5} - \tan^{-1}\frac{77}{36} = 0, 0 < \alpha < 13$, then $\sin^{-1}(\sin\alpha) + \cos^{-1}(\cos\alpha)$ is equal to :

(A) 16 (B) π (C) $16 - 5\pi$ (D) 0

Open solution

Q28. MCQ (Single Correct Answer)

If the sum and product of four positive consecutive terms of a G.P., are 126 and 1296 , respectively, then the sum of common ratios of all such GPs is

(A) 7 (B) 14 (C) 3 (D) $\frac{9}{2}$

Open solution

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Q29. MCQ (Single Correct Answer)

Open solution

For the system of linear equations

$x + y + z = 6$

$\alpha x + \beta y + 7z = 3$

$x + 2y + 3z = 14$

which of the following is **NOT** true ?

(A) If $\alpha = \beta = 7$, then the system has no solution

(B) For every point $(\alpha, \beta) \neq (7, 7)$ on the line $x - 2y + 7 = 0$, the system has infinitely many solutions

(C) There is a unique point (α, β) on the line $x + 2y + 18 = 0$ for which the system has infinitely many solutions

(D) If $\alpha = \beta$ and $\alpha \neq 7$, then the system has a unique solution

Q30. MCQ (Single Correct Answer)

Open solution

Let R be a relation on $\mathbb{N} \times \mathbb{N}$ defined by $(a, b)\ \mathrm{R}\ (c, d)$ if and only if $ad(b - c) = bc(a - d)$. Then R is

(A) symmetric and transitive but not reflexive

(B) reflexive and symmetric but not transitive

(C) transitive but neither reflexive nor symmetric

(D) symmetric but neither reflexive nor transitive

Q31. MCQ (Single Correct Answer)

Open solution

If the domain of the function $f(x) = \frac{[x]}{1+x^2}$, where $[x]$ is greatest integer $\leq x$, is $[2, 6)$, then its range is

(A) $\left(\frac{5}{37}, \frac{2}{5}\right] - \left\{\frac{9}{29}, \frac{27}{109}, \frac{18}{89}, \frac{9}{53}\right\}$

(B) $\left(\frac{5}{37}, \frac{2}{5}\right]$

(C) $\left(\frac{5}{26}, \frac{2}{5}\right]$

(D) $\left(\frac{5}{26}, \frac{2}{5}\right] - \left\{\frac{9}{29}, \frac{27}{109}, \frac{18}{89}, \frac{9}{53}\right\}$

Q32. MCQ (Single Correct Answer)

Open solution

A wire of length $20\ \mathrm{m}$ is to be cut into two pieces. A piece of length l_1 is bent to make a square of area A_1 and the other piece of length l_2 is made into a circle of area A_2. If $2A_1 + 3A_2$ is minimum then $(\pi l_1) : l_2$ is equal to :

(A) 6 : 1 (B) 1 : 6 (C) 4 : 1 (D) 3 : 1

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Q33. MCQ (Single Correct Answer)

Let $\alpha \in (0,1)$ and $\beta = \log_e(1-\alpha)$. Let $P_n(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \ldots + \frac{x^n}{n}, x \in (0,1)$. Then the integral $\int_0^{\alpha} \frac{t^{50}}{1-t} dt$ is equal to

Open solution

(A) $-(\beta + P_{50}(\alpha))$

(B) $\beta - P_{50}(\alpha)$

(C) $P_{50}(\alpha) - \beta$

(D) $\beta + P_{50}(\alpha)$

Q34. MCQ (Single Correct Answer)

Let the shortest distance between the lines

$L : \frac{x-5}{-2} = \frac{y-\lambda}{0} = \frac{z+\lambda}{1}, \lambda \geq 0$ and

$L_1 : x + 1 = y - 1 = 4 - z$ be $2\sqrt{6}$. If (α, β, γ) lies on L,

then which of the following is NOT possible?

Open solution

(A) $\alpha + 2\gamma = 24$

(B) $2\alpha + \gamma = 7$

(C) $\alpha - 2\gamma = 19$

(D) $2\alpha - \gamma = 9$

Q35. MCQ (Single Correct Answer)

Let $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$, and $\vec{b}$ and $\vec{c}$ be two nonzero vectors such that $|\vec{a} + \vec{b} + \vec{c}| = |\vec{a} + \vec{b} - \vec{c}|$ and $\vec{b} \cdot \vec{c} = 0$. Consider the following two statements:

Open solution

(A) $|\vec{a} + \lambda\vec{c}| \geq |\vec{a}|$ for all $\lambda \in \mathbb{R}$.

(B) $\vec{a}$ and $\vec{c}$ are always parallel.

Then,

(A) only (B) is correct

(B) both (A) and (B) are correct

(C) only (A) is correct

(D) neither (A) nor (B) is correct

Q36. MCQ (Single Correct Answer)

Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{pmatrix}$. Then the sum of the diagonal elements of the matrix $(A + I)^{11}$ is equal to :

Open solution

(A) 4094

(B) 2050

(C) 6144

(D) 4097

Q37. MCQ (Single Correct Answer)

For all $z \in C$ on the curve $C_1 : |z| = 4$, let the locus of the point $z + \frac{1}{z}$ be the curve C_2. Then :

Open solution

(A) the curves C_1 and C_2 intersect at 4 points

(B) the curve C_2 lies inside C_1

(C) the curve C_1 lies inside C_2

(D) the curves C_1 and C_2 intersect at 2 points

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Q38. MCQ (Single Correct Answer)

The value of $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{(2+3\sin x)}{\sin x(1+\cos x)}dx$ is equal to :

(A) $\frac{10}{3} - \sqrt{3} + \log_e \sqrt{3}$

(B) $\frac{7}{2} - \sqrt{3} - \log_e \sqrt{3}$

(C) $\frac{10}{3} - \sqrt{3} - \log_e \sqrt{3}$

(D) $-2 + 3\sqrt{3} + \log_e \sqrt{3}$

Open solution

Q39. MCQ (Single Correct Answer)

Let $y = f(x) = \sin^3\left(\frac{\pi}{3}\left(\cos\left(\frac{\pi}{3\sqrt{2}}\left(-4x^3+5x^2+1\right)^{\frac{3}{2}}\right)\right)\right)$. Then, at x = 1,

(A) $2y' + \sqrt{3}\pi^2 y = 0$

(B) $y' + 3\pi^2 y = 0$

(C) $\sqrt{2}y' - 3\pi^2 y = 0$

(D) $2y' + 3\pi^2 y = 0$

Open solution

Q40. Numerical

The remainder on dividing 5^{99} by 11 is _____.

Open solution

Q41. Numerical

Let 5 digit numbers be constructed using the digits $0, 2, 3, 4, 7, 9$ with repetition allowed, and are arranged in ascending order with serial numbers. Then the serial number of the number 42923 is _____.

Open solution

Q42. Numerical

Let $\alpha > 0$, be the smallest number such that the expansion of $\left(x^{\frac{2}{3}} + \frac{2}{x^3}\right)^{30}$ has a term $\beta x^{-\alpha}, \beta \in \mathbb{N}$. Then α is equal to _____.

Open solution

Q43. Numerical

Number of 4-digit numbers that are less than or equal to 2800 and either divisible by 3 or by 11 , is equal to _____.

Open solution

Q44. Numerical

Let $a_1, a_2, \ldots, a_n$ be in A.P. If $a_5 = 2a_7$ and $a_{11} = 18$, then

$12\left(\frac{1}{\sqrt{a_{10}}+\sqrt{a_{11}}} + \frac{1}{\sqrt{a_{11}}+\sqrt{a_{12}}} + \ldots + \frac{1}{\sqrt{a_{17}}+\sqrt{a_{18}}}\right)$ is equal to _____.

Open solution

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Q45. Numerical

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}| = \sqrt{14}$, $|\vec{b}| = \sqrt{6}$ and $|\vec{a} \times \vec{b}| = \sqrt{48}$. Then $(\vec{a} \cdot \vec{b})^2$ is equal to _____.

Open solution

Q46. Numerical

Let for $x \in \mathbb{R}$,

$f(x) = \frac{x+|x|}{2}$ and $g(x) = \begin{cases} x, & x < 0 \\ x^2, & x \geq 0 \end{cases}$.

Then area bounded by the curve $y = (f \circ g)(x)$ and the lines $y = 0, 2y - x = 15$ is equal to _____.

Open solution

Q47. Numerical

If the variance of the frequency distribution

x_i	2	3	4	5	6	7	8
Frequency f_i	3	6	16	α	9	5	6

is 3, then α is equal to _____.

Open solution

Physics 27 questions

Q48. MCQ (Single Correct Answer)

The initial speed of a projectile fired from ground is u. At the highest point during its motion, the speed of projectile is $\frac{\sqrt{3}}{2}u$. The time of flight of the projectile is :

(A) $\frac{u}{g}$ (B) $\frac{2u}{g}$ (C) $\frac{u}{2g}$ (D) $\frac{\sqrt{3}u}{g}$

Open solution

Q49. MCQ (Single Correct Answer)

A bar magnet with a magnetic moment $5.0\,\mathrm{Am}^2$ is placed in parallel position relative to a magnetic field of $0.4\,\mathrm{T}$. The amount of required work done in turning the magnet from parallel to antiparallel position relative to the field direction is _____.

(A) zero (B) 1 J (C) 2 J (D) 4 J

Open solution

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Q50. MCQ (Single Correct Answer)

Spherical insulating ball and a spherical metallic ball of same size and mass are dropped from the same height. Choose the correct statement out of the following

{Assume negligible air friction}

Open solution

(A) Metal ball will reach the earth's surface earlier than the insulating ball

(B) Both will reach the earth's surface simultaneously.

(C) Insulating ball will reach the earth's surface earlier than the metal ball

(D) Time taken by them to reach the earth's surface will be independent of the properties of their materials

Q51. MCQ (Single Correct Answer)

The effect of increase in temperature on the number of electrons in conduction band (n_e) and resistance of a semiconductor will be as:

Open solution

(A) n_e decreases, resistance increases

(B) Both n_e and resistance increase

(C) n_e increases, resistance decreases

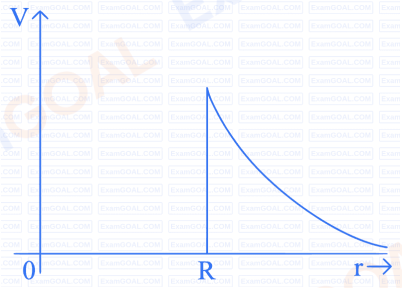
(D) Both n_e and resistance decrease

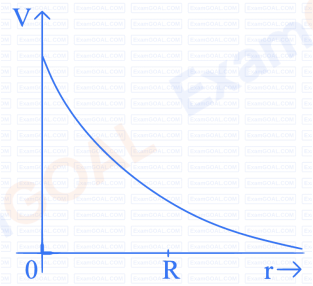
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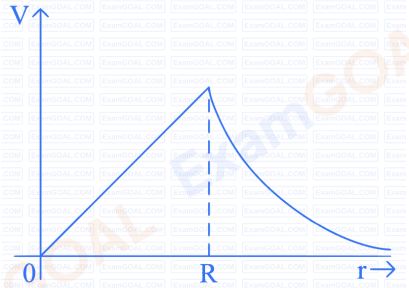
Q52. MCQ (Single Correct Answer)

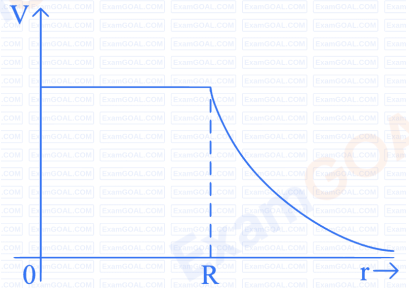
Which of the following correctly represents the variation of electric potential (V) of a charged spherical conductor of radius (R) with radial distance (r) from the center?

Open solution

(A) 

(B) 

(C) 

(D) 

Q53. MCQ (Single Correct Answer)

The drift velocity of electrons for a conductor connected in an electrical circuit is V_d. The conductor is now replaced by another conductor with same material and same length but double the area of cross section. The applied voltage remains same. The new drift velocity of electrons will be

Open solution

(A) $\frac{V_d}{4}$ (B) V_d (C) $2\,V_d$ (D) $\frac{V_d}{2}$

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Q54. MCQ (Single Correct Answer)

A rod with circular cross-section area $2\ \mathrm{cm}^2$ and length $40\ \mathrm{cm}$ is wound uniformly with 400 turns of an insulated wire. If a current of $0.4\ \mathrm{A}$ flows in the wire windings, the total magnetic flux produced inside windings is $4\pi \times 10^{-6}\ \mathrm{Wb}$. The relative permeability of the rod is

(Given : Permeability of vacuum $\mu_0 = 4\pi \times 10^{-7}\mathrm{NA}^{-2}$)

(A) $\frac{5}{16}$ (B) 125 (C) $\frac{32}{5}$ (D) 12.5

Open solution

Q55. MCQ (Single Correct Answer)

At a certain depth "d " below surface of earth, value of acceleration due to gravity becomes four times that of its value at a height $3\mathrm{R}$ above earth surface. Where R is Radius of earth (Take $\mathrm{R} = 6400\ \mathrm{km}$). The depth d is equal to

(A) 5260 km (B) 2560 km (C) 640 km (D) 4800 km

Open solution

Q56. MCQ (Single Correct Answer)

Two polaroids A and B are placed in such a way that the pass-axis of polaroids are perpendicular to each other. Now, another polaroid C is placed between A and B bisecting angle between them. If intensity of unpolarized light is I_0 then intensity of transmitted light after passing through polaroid B will be:

(A) $\frac{I_0}{4}$ (B) $\frac{I_0}{8}$ (C) Zero (D) $\frac{I_0}{2}$

Open solution

Q57. MCQ (Single Correct Answer)

100 balls each of mass m moving with speed v simultaneously strike a wall normally and reflected back with same speed, in time $\mathrm{t\ s}$. The total force exerted by the balls on the wall is

(A) $\frac{200mv}{t}$ (B) $\frac{100mv}{t}$ (C) $\frac{mv}{100t}$ (D) $200mvt$

Open solution

Q58. MCQ (Single Correct Answer)

If $\mathrm{R}, \mathrm{X_L}$, and $\mathrm{X_C}$ represent resistance, inductive reactance and capacitive reactance. Then which of the following is dimensionless :

(A) $\frac{R}{X_L X_C}$ (B) RX_LX_C (C) $\frac{R}{\sqrt{X_L X_C}}$ (D) $R\frac{X_L}{X_C}$

Open solution

Q59. MCQ (Single Correct Answer)

The maximum potential energy of a block executing simple harmonic motion is $25\ \mathrm{J}$. A is amplitude of oscillation. At $\mathrm{A}/2$, the kinetic energy of the block is

(A) 9.75 J (B) 37.5 J (C) 18.75 J (D) 12.5 J

Open solution

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Q60. MCQ (Single Correct Answer)

If 1000 droplets of water of surface tension $0.07\ \mathrm{N/m}$, having same radius $1\ \mathrm{mm}$ each, combine to form a single drop. In the process the released surface energy is :-

$\left(\text{Take } \pi = \frac{22}{7}\right)$

Open solution

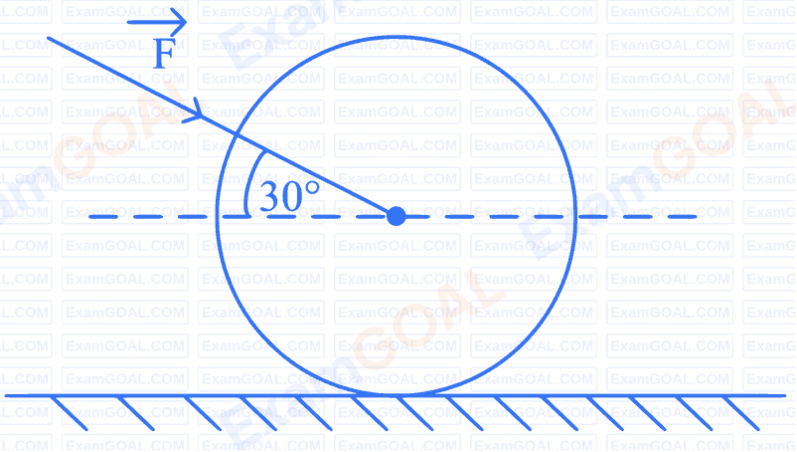
(A) $7.92 \times 10^{-4}\ \mathrm{J}$ (B) $7.92 \times 10^{-6}\ \mathrm{J}$ (C) $8.8 \times 10^{-5}\ \mathrm{J}$ (D) $9.68 \times 10^{-4}\ \mathrm{J}$

Q61. MCQ (Single Correct Answer)

As shown in figure, a $70\ \mathrm{kg}$ garden roller is pushed with a force of $\vec{F} = 200\ \mathrm{N}$ at an angle of 30° with horizontal. The normal reaction on the roller is

(Given $g = 10\ \mathrm{m\,s^{-2}}$)

Open solution

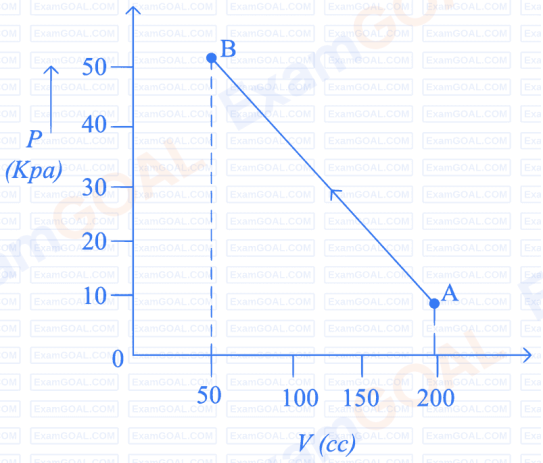


(A) 800 N (B) 600 N (C) $200\sqrt{3}$ N (D) $800\sqrt{2}$ N

Q62. MCQ (Single Correct Answer)

The pressure of a gas changes linearly with volume from A to B as shown in figure. If no heat is supplied to or extracted from the gas then change in the internal energy of the gas will be

Open solution



(A) 6 J (B) 4.5 J (C) zero (D) -4.5 J

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Q63. MCQ (Single Correct Answer)

The correct relation between $\gamma = \frac{c_p}{c_v}$ and temperature T is :

(A) $\gamma \propto T$

(B) $\gamma \propto \frac{1}{\sqrt{T}}$

(C) $\gamma \propto \frac{1}{T}$

(D) $\gamma \propto T^0$

Open solution

Q64. MCQ (Single Correct Answer)

If a source of electromagnetic radiation having power $15\,\text{kW}$ produces 10^{16} photons per second, the radiation belongs to a part of spectrum is.

(Take Planck constant $h = 6 \times 10^{-34}\text{Js}$)

(A) Gamma rays (B) Radio waves (C) Micro waves (D) Ultraviolet rays

Open solution

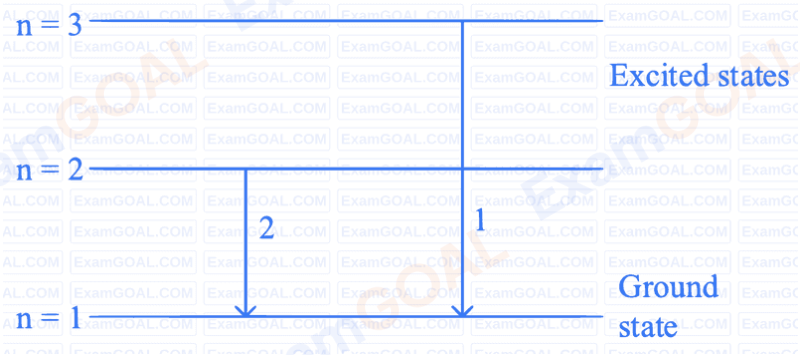
Q65. Numerical

A lift of mass $\text{M} = 500\,\text{kg}$ is descending with speed of $2\,\text{ms}^{-1}$. Its supporting cable begins to slip thus allowing it to fall with a constant acceleration of $2\,\text{ms}^{-2}$. The kinetic energy of the lift at the end of fall through to a distance of $6\,\text{m}$ will be _____ kJ.

Open solution

Q66. Numerical

For hydrogen atom, λ_1 and λ_2 are the wavelengths corresponding to the transitions 1 and 2 respectively as shown in figure. The ratio of λ_1 and λ_2 is $\frac{x}{32}$. The value of x is _____.

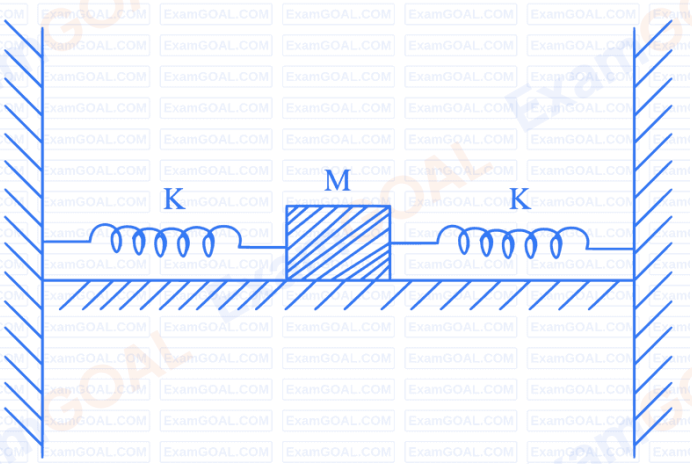


Open solution

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Q67. Numerical

In the figure given below, a block of mass $M = 490\ \mathrm{g}$ placed on a frictionless table is connected with two springs having same spring constant $\left(\mathrm{K} = 2\ \mathrm{N\ m^{-1}}\right)$. If the block is horizontally displaced through 'X' m then the number of complete oscillations it will make in 14π seconds will be _____.



Open solution

Q68. Numerical

The speed of a swimmer is $4\ \mathrm{km\ h^{-1}}$ in still water. If the swimmer makes his strokes normal to the flow of river of width $1\ \mathrm{km}$, he reaches a point $750\ \mathrm{m}$ down the stream on the opposite bank.

The speed of the river water is _____ $\mathrm{km\ h^{-1}}$

Open solution

Q69. Numerical

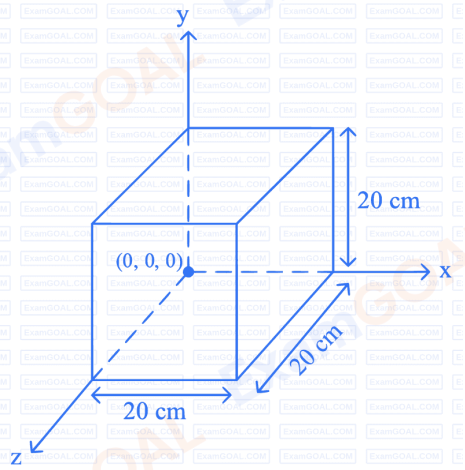
A solid sphere of mass $1\ \mathrm{kg}$ rolls without slipping on a plane surface. Its kinetic energy is $7 \times 10^{-3}\ \mathrm{J}$. The speed of the centre of mass of the sphere is _____ $\mathrm{cm\ s^{-1}}$

Open solution

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Q70. Numerical

Expression for an electric field is given by $\overrightarrow{E} = 4000x^2\hat{i}\,\frac{V}{m}$. The electric flux through the cube of side $20\ cm$ when placed in electric field (as shown in the figure) is _____ $V\ cm$.



Open solution

Q71. Numerical

A thin rod having a length of $1\ m$ and area of cross-section $3 \times 10^{-6}\ m^2$ is suspended vertically from one end. The rod is cooled from $210°C$ to $160°C$. After cooling, a mass M is attached at the lower end of the rod such that the length of rod again becomes $1\ m$. Young's modulus and coefficient of linear expansion of the rod are $2 \times 10^{11}\ N\,m^{-2}$ and $2 \times 10^{-5}\ K^{-1}$, respectively. The value of M is _____ kg.

(Take $g = 10\ m\,s^{-2}$)

Open solution

Q72. Numerical

In a medium the speed of light wave decreases to 0.2 times to its speed in free space. The ratio of relative permittivity to the refractive index of the medium is $x : 1$. The value of x is _____.

(Given speed of light in free space $= 3 \times 10^8\ m\,s^{-1}$ and for the given medium $\mu_r = 1$)

Open solution

Q73. Numerical

Two identical cells, when connected either in parallel or in series gives same current in an external resistance $5\ \Omega$. The internal resistance of each cell will be _____ Ω.

Open solution

Q74. Numerical

An inductor of $0.5\ mH$, a capacitor of $20\ \mu F$ and resistance of $20\ \Omega$ are connected in series with a $220\ V$ ac source. If the current is in phase with the emf, the amplitude of current of the circuit is $\sqrt{x}$ A. The value of x is _____

Open solution

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Out of syllabus 16 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q75. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The methods NOT involved in concentration of ore are

A. Liquation

B. Leaching

C. Electrolysis

D. Hydraulic washing

E. Froth floatation

Choose the correct answer from the options given below :

(A) A and C only **(B)** C, D and E only **(C)** B, D and C only **(D)** B, D and E only

Q76. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Identify X, Y and Z in the following reaction. (Equation not balanced)

$$ClO^{\bullet} + NO_2 \rightarrow \underline{X} \xrightarrow{H_2O} \underline{Y} + \underline{Z}$$

(A) $X = ClONO_2, Y = HOCl, Z = NO_2$

(B) $X = ClNO_3, Y = Cl_2, Z = NO_2$

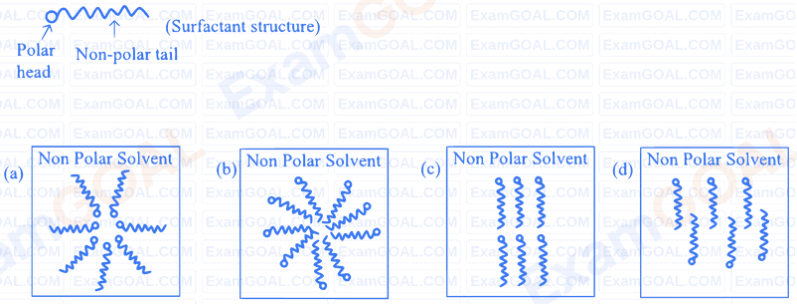
(C) $X = ClONO_2, Y = HOCl, Z = HNO_3$

(D) $X = ClNO_2, Y = HCl, Z = HNO_3$

Q77. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Adding surfactants in non polar solvent, the micelles structure will look like



(A) d **(B)** a **(C)** c **(D)** b

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Q78. MCQ (Single Correct Answer) · **Chemistry**

H_2O_2 acts as a reducing agent in :

(A) $Mn^{2+} + 2H_2O_2 \rightarrow MnO_2 + 2H_2O$

(B) $2Fe^{2+} + 2H^+ + H_2O_2 \rightarrow 2Fe^{3+} + 2H_2O$

(C) $Na_2S + 4H_2O_2 \rightarrow Na_2SO_4 + 4H_2O$

(D) $2NaOCl + H_2O_2 \rightarrow 2NaCl + H_2O + O_2$

Open solution

Q79. MCQ (Single Correct Answer) · **Chemistry**

Which of the following artificial sweeteners has the highest sweetness value in comparison to cane sugar ?

(A) Alitame (B) Sucralose (C) Saccharin (D) Aspartame

Open solution

Q80. Numerical · **Chemistry**

The total pressure of a mixture of non-reacting gases $X(0.6\ g)$ and $Y(0.45\ g)$ in a vessel is $740\ mm$ of Hg.

The partial pressure of the gas X is _____ mm of Hg. (Nearest Integer)

(Given : molar mass $X = 20$ and $Y = 45\ g\ mol^{-1}$)

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

Let a differentiable function f satisfy $f(x) + \int_3^x \frac{f(t)}{t}dt = \sqrt{x+1}, x \geq 3$. Then $12f(8)$ is equal to :

(A) 19 (B) 34 (C) 17 (D) 1

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Let a circle C_1 be obtained on rolling the circle $x^2 + y^2 - 4x - 6y + 11 = 0$ upwards 4 units on the tangent T to it at the point $(3, 2)$. Let C_2 be the image of C_1 in T. Let A and B be the centers of circles C_1 and C_2 respectively, and M and N be respectively the feet of perpendiculars drawn from A and B on the x-axis. Then the area of the trapezium AMNB is :

(A) $2\left(2+\sqrt{2}\right)$ (B) $4\left(1+\sqrt{2}\right)$ (C) $3+2\sqrt{2}$ (D) $2\left(1+\sqrt{2}\right)$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

If the maximum distance of normal to the ellipse $\frac{x^2}{4} + \frac{y^2}{b^2} = 1, b < 2$, from the origin is 1, then the eccentricity of the ellipse is :

(A) $\frac{\sqrt{3}}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{2}$

Open solution

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Q84. MCQ (Single Correct Answer) · **Mathematics**

Open solution

$(S1)$ $(p \Rightarrow q) \vee (p \wedge (\sim q))$ is a tautology

$(S2)$ $((\sim p) \Rightarrow (\sim q)) \wedge ((\sim p) \vee q)$ is a contradiction.

Then

(A) only (S2) is correct

(B) both (S1) and (S2) are correct

(C) only (S1) is correct

(D) both (S1) and (S2) are wrong

Q85. MCQ (Single Correct Answer) · **Mathematics**

Open solution

Let $y = f(x)$ represent a parabola with focus $\left(-\frac{1}{2}, 0\right)$ and directrix $y = -\frac{1}{2}$. Then

$$S = \left\{x \in \mathbb{R} : \tan^{-1}(\sqrt{f(x)}) + \sin^{-1}(\sqrt{f(x)+1}) = \frac{\pi}{2}\right\} :$$

(A) is an empty set

(B) contains exactly one element

(C) contains exactly two elements

(D) is an infinite set

Q86. Numerical · **Mathematics**

Open solution

Let the line $L : \frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-3}{1}$ intersect the plane $2x + y + 3z = 16$ at the point P. Let the point Q be the foot of perpendicular from the point $R(1, -1, -3)$ on the line L. If α is the area of triangle PQR, then α^2 is equal to _____.

Q87. Numerical · **Mathematics**

Open solution

Let θ be the angle between the planes $P_1 : \vec{r} \cdot (\hat{i} + \hat{j} + 2\hat{k}) = 9$ and $P_2 : \vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) = 15$. Let L be the line that meets P_2 at the point $(4, -2, 5)$ and makes an angle θ with the normal of P_2. If α is the angle between L and P_2, then $\left(\tan^2\theta\right)\left(\cot^2\alpha\right)$ is equal to _____.

Q88. MCQ (Single Correct Answer) · **Physics**

Open solution

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R

Assertion A : The beam of electrons show wave nature and exhibit interference and diffraction.

Reason R : Davisson Germer Experimentally verified the wave nature of electrons.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) A is not correct but R is correct.

(B) Both A and R are correct and R is the correct explanation of A

(C) Both A and R are correct but R is Not the correct explanation of A

(D) A is correct but R is not correct

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Q89. MCQ (Single Correct Answer) · **Physics**

The amplitude of $15\sin(1000\pi t)$ is modulated by $10\sin(4\pi t)$ signal. The amplitude modulated signal contains frequency (ies) of

A. 500 Hz

B. 2 Hz

C. 250 Hz

D. 498 Hz

E. 502 Hz

Choose the correct answer from the options given below:

(A) B only (B) A only (C) A, D and E only (D) A and B only

Open solution

Q90. MCQ (Single Correct Answer) · **Physics**

A free neutron decays into a proton but a free proton does not decay into neutron. This is because

(A) neutron is an uncharged particle

(B) neutron has larger rest mass than proton

(C) neutron is a composite particle made of a proton and an electron

(D) proton is a charged particle

Open solution

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Answer Key

Chemistry

1. **B**	2. **D**	3. **C**	4. **C**	5. **B**
6. **C**	7. **A**	8. **A**	9. **C**	10. **A**
11. **B**	12. **D**	13. **A**	14. **D**	15. **B**
16. **4**	17. **6**	18. **610**	19. **44**	20. **62250**
21. **4**	22. **3**	23. **2520**	24. **1**	

Mathematics

25. **B**	26. **C**	27. **B**	28. **A**	29. **B**
30. **D**	31. **B**	32. **A**	33. **A**	34. **A**
35. **C**	36. **D**	37. **A**	38. **A**	39. **D**
40. **9**	41. **2997**	42. **2**	43. **710**	44. **8**
45. **36**	46. **72**	47. **5**		

Physics

48. **A**	49. **D**	50. **C**	51. **C**	52. **D**
53. **B**	54. **A**	55. **D**	56. **B**	57. **A**
58. **C**	59. **C**	60. **A**	61. **A**	62. **B**
63. **D**	64. **A**	65. **7**	66. **27**	67. **20**
68. **3**	69. **10**	70. **640**	71. **60**	72. **5**
73. **5**	74. **242**			

Out of syllabus

75. **A**	76. **C**	77. **B**	78. **D**	79. **A**
80. **555**	81. **C**	82. **B**	83. **D**	84. **C**
85. **C**	86. **180**	87. **9**	88. **B**	89. **C**
90. **B**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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